

STANDARD = Variational autoencoder (VAE)

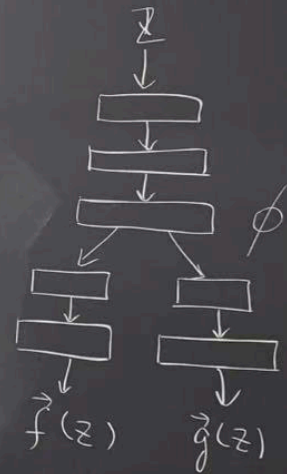


$$z \sim N(\vec{0}, I) = p(z)$$



$$x \sim N(x | \vec{f}(z), \text{diag}(\vec{g}(z)))$$

$$p_{\phi}(z, x) = p(z) p_{\phi}(x|z) \quad p(x|z)$$



① Train

② Gener



③ Unders

$$p(z)$$

① Train the model ✓.

② Generate data.

$$\begin{array}{l} \textcircled{z} \quad \hat{z} \sim N(\vec{\mu}, \mathbf{I}) \\ \downarrow \\ \textcircled{x} \quad \hat{x} \sim N(x | \vec{f}(\hat{z}), \text{diag}(\vec{g}(\hat{z}))) \end{array}$$

③ Understand data

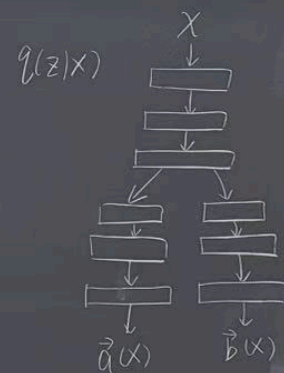
$p(z|x)$ inference

Variational inference

$$p(z|x) \approx q(z|x)$$

$$\min_{\theta} \text{KL}(q_{\theta}(z|x) \| p(z|x))$$

$$\theta^* \quad \underline{q_{\theta^*}(z|x)}$$



$$q_{\theta}(z|x) = N(z | \vec{a}(x), \vec{b}(x))$$

Maximum likelihood estimation (MLE) ✓

Moment matching X

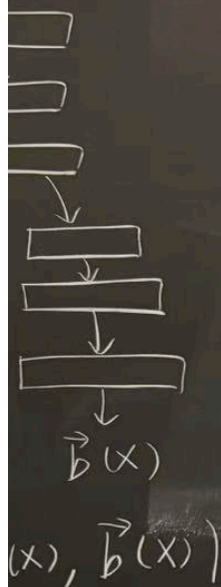
$$P_{\theta}(x) \quad \frac{x_1 \ x_2 \ x_3 \ \dots \ x_N}{}$$

$$\log \prod_{i=1}^N P_{\theta}(x_i) = \sum_{i=1}^N \log P_{\theta}(x_i)$$

data likelihood log data likelihood

$$\max_{\theta} \sum_{i=1}^N \log P_{\theta}(x_i)$$

θ^*





Real images $X_1, X_2, \dots, X_N$ autoencoder (VAE)

$$\max_{\phi} \sum_{i=1}^N \log P_{\phi}(X_i)$$

$$\log P_{\phi}(x)$$

$$= \log \int_z P(x, z) dz$$

sum rule

$$= \log \int_z \frac{P_{\phi}(x, z)}{q_{\theta}(z|x)} q_{\theta}(z|x) dz$$

$$= \log E_{q_{\theta}(z|x)} \left[\frac{P_{\phi}(x, z)}{q_{\theta}(z|x)} \right]$$

$$\geq E_{q_{\theta}(z|x)} \left[\log \frac{P_{\phi}(x, z)}{q_{\theta}(z|x)} \right]$$

$$= E_{q_{\theta}(z|x)} \left[\log P_{\phi}(x, z) \right] - E_{q_{\theta}(z|x)} \left[\log q_{\theta}(z|x) \right]$$

Evidence lower-bound (ELBO)

Jensen's

If a function

then

$f(E[X])$

$$\int \frac{q_\theta(z|x)}{q_\theta(z|x)} dz$$

$$\log \frac{p_\phi(x, z)}{q_\theta(z|x)}$$

$$\log \frac{p_\phi(x, z)}{q_\theta(z|x)}$$

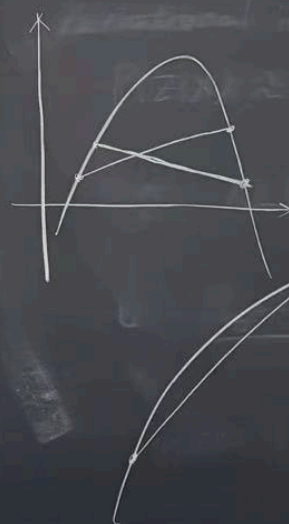
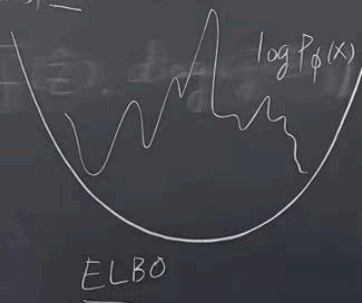
$$\log p_\phi(x, z) - E_{q_\theta(z|x)} [\log q_\theta(z|x)]$$

evidence lower-bound (ELBO)

Jensen's inequality.

If a function f is concave,
then

$$f(E[x]) \geq E[f(x)]$$



$$\max_{\phi, \theta} \sum_{i=1}^N E_{q_\theta(z_i|x_i)} [\log p_\phi(x_i, z_i)]$$

ϕ : model parameters
 θ : auxiliary parameters

$$\max_{\phi, \theta} \sum_{i=1}^N E_{q_{\theta}(z_i | x_i)} [f(\phi, \theta) \log p_{\phi}(x_i, z_i)]$$

ϕ : model parameters ✓
 θ : auxiliary parameters

Maximum likelihood estimation (MLE) ✓
 $E_{q_{\theta}(z_i | x_i)} [\log q_{\theta}(z_i | x_i)]$

While not converged

- ① fix ϕ , optimize θ — Expectation step
- ② fix θ , optimize ϕ — Maximization step

Coordinate ascent
 Expectation maximization (EM)

$$q(z|x) = \prod_{i=1}^N q(z_i | x_i)$$

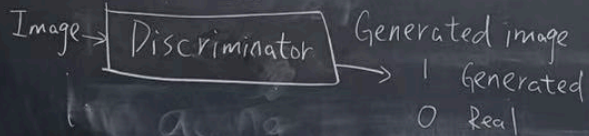


Generative adversarial network (GAN)

Human
tester



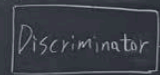
tester



Real
Training
image

choose

Real image



Real

Fake

Generator

Generated image

fool



$$\hat{z} \sim N(\vec{0}, I)$$

$$\hat{x} \sim N(x | f(\hat{z}), \text{diag}(g(\hat{z})))$$

Jensen's inequality.
If a function f is convex then
 $f(E[x]) \leq E[f(x)]$

ELBO